Abikan Hospital & Wellness Centre

Abikan EMR

Staff and operations documentation

Release **v1.0.0-uat**

Commit **2ea80c40c02cd42aa33fba47a0aa9dc1cc7d9ff6**

This PDF describes the EMR as it exists in this release. It is not a brochure and does not list planned features as if they were available.

Do not use development passwords on the hospital server. Do not share accounts.

Abikan EMR — UAT quick reference

Print this page. Release **v1.0.0-uat**. Use **synthetic** patients only. Do not use development passwords on the hospital server.

Department	Primary responsibility	Main workflow
Reception	Registration and visits	Search → register if new → start visit → card fee → send to Clinical. Pay at Accountant
Doctor	Consultation, diagnosis, orders	Today's Patients → consult → diagnose / order / prescribe → Admit Patient if needed
Nursing	Ward, vitals, notes, MAR	Ward Board → assign bed → notes / vitals / scheduled MAR → discharge
Laboratory	Orders and results visibility	See order → process in laboratory processing → result returns to Doctor
Pharmacy	Dispense after payment	Paid → Ready to dispense → Dispense. Do not administer MAR
Accountant	Collect payment	Search → collect → receipt. Billed / collected / outstanding are bill totals, not “revenue”
Radiology	Price and report	Worklist → set naira price → write report. Images not in Abikan
Hospital Admin	Staff and counts	Staff accounts · overview · staff activity. Occupancy and outstanding are **now**
Inpatient	Shared ward path	Doctor admits → Nurse beds and documents → Accountant collects accommodation

Safety

- Confirm name and hospital number every time.
- Do not share accounts.
- Do not change times (vitals and MAR are automatic).
- Do not dispense unpaid medication.
- Do not enter laboratory results in Abikan.
- Pharmacy dispenses. Nursing administers scheduled inpatient doses. PRN is not on MAR.

Money

New bills: **naira**. Old accommodation bills in dollars stay in dollars. Changing a price does not change an old bill.

Known UAT limits

- Prove a **new** laboratory result here.
- Laboratory board can be slow.
- Nursing Diagnoses may be unavailable; tasks / care plan / intake show Coming soon.
- SuperAdmin can change catalogue prices and accommodation tariffs.
- Unconfigured ward tariff may be **0**.
- System Settings, doctor discharge decision, refunds, and radiology images: **not currently available in this release**.

Full manuals: start at ABIKAN_EMR_USER_MANUAL.md. Sign-off: HOSPITAL_UAT_CHECKLIST.md.

Abikan EMR — User Manual

This manual describes **Abikan EMR as it exists in the frozen hospital UAT release** (`v1.0.0-uat`). It is for hospital staff and trainers.

It is not a product brochure. If something is not available in this release, this package says so.

Release: `v1.0.0-uat` · commit `2ea80c40c02cd42aa33fba47a0aa9dc1cc7d9ff6`

What is Abikan EMR?

Abikan EMR is the hospital's daily work system for Abikan Hospital & Wellness Centre.

Staff sign in once and open **their department**. They register patients, see today's visits, record consultations, order tests and medicines, nurse admitted patients, dispense paid prescriptions, collect payment, and review hospital activity.

Abikan is the **staff-facing workflow layer**. It does not replace the hospital's clinical and financial records. Those remain in the systems already used for the chart, laboratory processing, billing, and imaging.

What staff use	What it is for
Abikan	Department screens, hospital language, daily steps, department pricing, cashier, ward board, MAR, analytics
Clinical chart engine	Patient identity, visits, diagnoses, prescriptions, observations, admissions
Laboratory processing	Specimen collection, accession, testing, result entry, validation
Billing and stock	Invoices, payments, receipts, pharmacy stock movements
Imaging archive	Scan files, when a modality is connected. **Images are not shown inside Abikan in this release.**

Staff should think in hospital tasks: register, consult, collect specimen, dispense, collect payment. They should not need to know the names of the engines behind the screens.

Who uses it

Department (staff name)	Typical work in Abikan
Reception	Find or register a patient, start a visit, bill a registration card, set card prices
Clinical (Doctor)	Today's patients, consultation, diagnosis, laboratory and imaging orders, prescriptions, admit decision
Nursing	Ward board, assign / transfer / discharge, nursing notes, inpatient vitals, medication administration
Laboratory	See ordered tests, payment state, open laboratory processing, set test prices
Pharmacy	See prescriptions, dispense after payment, catalogue and drug prices, inventory when permitted
Accountant	Find a patient, collect payment, print or open a receipt
Radiology	Worklist, set an examination price, write a report
Hospital Admin	Staff accounts, department access, hospital overview, staff activity
SuperAdmin	Technical hospital-wide access, including accommodation prices and every department

An account only opens the departments assigned to it. Signing in does not let a nurse collect payment or a doctor change laboratory prices.

How the hospital works in Abikan

Clinical workflow

1. Reception finds or registers the patient and starts a visit.
2. The doctor opens that visit, records the consultation, and may diagnose, order tests or imaging, prescribe, or request admission.
3. Laboratory, Pharmacy, and Radiology work from those orders.
4. The doctor reviews results and reports on the same visit.
5. If admitted, Nursing assigns a bed and records notes, vitals, and scheduled medicines.
6. Nursing discharges the patient and frees the bed.
7. The visit and records remain available for later care.

Payment is **not** required before the doctor can save a consultation or place an order. Unpaid work is visible. Collection happens at **Accountant**.

Financial workflow

1. A department creates a **charge** (registration card, consultation fee, laboratory fee, imaging price, pharmacy prescription, accommodation).
2. That charge becomes a bill the cashier can collect.
3. Accountant records **payment**.
4. The department screen then shows **Paid** (or pharmacy **Ready to dispense**).
5. Completing the service is a separate step (dispense, process a specimen, write a report).

Paid does not mean the work is finished.

Changing today's catalogue price does **not** change a bill already created.

New departmental bills use **naira (₦)**. The financial company currency may still be US dollars. Older accommodation bills that were in dollars stay in dollars. See BILLING_AND_PAYMENTS.md.

Administrative workflow

Hospital Admin creates and disables staff accounts, assigns departments, and reviews hospital counts and staff-administration activity.

Hospital Admin does **not** process specimens, write clinical notes, or collect payment (unless that person also has those department rights).

System Settings is not available in this release.

How information moves (staff view)

```
Patient arrives
    → Reception: find or register, start visit, card fee
    → Clinical: consultation, orders, prescription, admit if needed
        → Laboratory: see order → process in laboratory processing → result returns
        → Pharmacy: wait for payment → dispense
        → Radiology: price and report (images not in Abikan)
    → Doctor reviews results and reports
    → Nursing: bed, notes, vitals, scheduled medicines
    → Nursing: discharge
    → Accountant: collect any unpaid bills
    → Records remain on the patient
```

Only steps that this release supports are listed. Details are in CLINICAL_WORKFLOWS.md.

Signing in

1. Open the hospital Abikan address on a supported browser.
2. Enter **your hospital username** and **your own password**.
3. Select **Sign in**.
4. Abikan opens **your department**. You do not pick a room or ward as a workplace.

Wrong username or password: **Could not sign in** — *Incorrect username or password.*

If the network cannot be reached: **Unable to connect** — try again. If that continues, tell Hospital Admin.

Do not share accounts. Do not use another person's password. Do not use development passwords on the hospital system.

Sign out when you leave the desk.

Finding a patient

Most departments use the header search:

Search by name, hospital number, or phone (type at least two characters).

Always confirm **name and hospital number** before you record, order, dispense, administer, or collect money.

Patient Account

Patient Account is a read-only money view: billed so far, already received, still unpaid, and department bills.

Reception, Clinical, Nursing, Laboratory, Pharmacy, Radiology, and Accountant can open it when they can already open that patient. **Only Accountant collects payment.**

Open full chart

Some department patient screens offer **Open full chart**. That opens the full clinical chart for extra detail.

Use it when you need the wider chart. Daily work stays in Abikan. Do not treat extra chart screens as Abikan department procedures.

Open full chart is not shown on the Doctor consultation screen in this release. It is available on Nursing, Laboratory, and Pharmacy patient screens.

What staff should never do

- Share a username or password
- Register a new patient without searching first
- Record work on the wrong hospital number
- Change times on vitals or medication administration (time is recorded automatically)
- Bypass payment to dispense (the system blocks unpaid dispensing)
- Enter or invent laboratory results in Abikan (results are entered in laboratory processing)
- Dispense a medicine from the ward MAR, or administer a dose from Pharmacy
- Delete or "fix" records in hidden systems
- Use development passwords on the hospital server

Known limitations (this release)

These are accepted UAT limits, not hidden defects.

- A **new** laboratory result should be processed during hospital UAT. Do not rely only on an older result from another environment.
- The laboratory board can take several seconds to load.
- New departmental bills use naira. The company currency may still be US dollars. Older accommodation invoices in dollars stay in dollars.
- Nursing **Diagnoses** may show as unavailable.
- Some Nursing sections show **Coming soon** (tasks, care plan, intake and output).
- SuperAdmin can still change department catalogue prices and accommodation tariffs.
- Wards without a configured accommodation tariff show **0** until a price is set.
- When-needed (PRN) medicines are **not** added to the scheduled medication sheet.
- Radiology **images are not shown in Abikan**.
- **System Settings** is not currently available in this release.
- Doctor **Refer**, **Follow-up disposition**, and **doctor discharge decision** are not currently available in this release. Nursing performs discharge from the ward.
- Examination, Review of Systems, Family History, and Social History are **not currently available** on the Doctor consultation screen.
- Cashier **refunds** are not currently available in this release.
- Four laboratory tests remain unordered from Abikan (Inorganic Phosphate, Testosterone, Progesterone, Estradiol) because they have no safe hospital catalogue mapping.

Department manuals

Manual	Audience
Reception	Front desk
Doctor	Clinical
Nursing	Ward
Inpatient / ward	Shared ward operations
Laboratory	Laboratory
Pharmacy	Pharmacy
Accountant	Cashier / accounts
Radiology	Imaging desk
Hospital Admin	Administration

Other guides in this package

Document	Use
UAT Quick Reference	One-page printout
Clinical workflows	Journey diagrams
Billing and payments	Charges, bills, payment
Roles and permissions	Who can do what
Troubleshooting	When something goes wrong
System architecture	How the stack fits together (technical)

Hospital install and UAT (already published)

Do not treat these as staff desk manuals. They remain the install and sign-off set.

- Hospital UAT checklist

- Hospital deployment
- Hospital server installation
- Deployment secrets (variable names only)
- Department pricing
- Hospital overview analytics
- Inpatient implementation notes

Reception — Department guide

This guide describes Reception as it works in this release. Payment is collected at Accountant, not at Reception.

Related: User manual · Billing · Department pricing

What Reception is for

Reception finds or registers the patient, starts the right kind of visit, bills the registration card, and sends the patient to Clinical, Emergency, or the ward as needed.

Screens: Dashboard · Search Patients · Register Patient · Today's Visits · Pricing · Patient Accounts (search)

Logging in

PURPOSE: Open the Reception desk.

WHO DOES IT: Staff with Reception or Registration access.

STEP-BY-STEP

1. Open Abikan and enter your hospital username and password.
2. Select **Sign in**.
3. You open **Reception**. You do not pick a room.

WHAT HAPPENS AFTER SAVING: Your session stays in Reception until you sign out or switch departments (only if your account has more than one).

COMMON ERRORS: *Could not sign in* — incorrect username or password.

WHAT TO DO IF SOMETHING FAILS: Try again. If it still fails, contact Hospital Admin. Do not use someone else's account.

SECURITY NOTES: Use only your own password. Sign out when you leave the desk.

Finding an existing patient

PURPOSE: Open the correct person. Avoid a second registration.

WHO DOES IT: Reception.

STEP-BY-STEP

1. Use **Search by name, hospital number, or phone** (type at least two characters).
2. Confirm **name and hospital number**.
3. Open the patient.

Search Patients and **Patient Accounts** focus the same search.

WHAT HAPPENS AFTER SAVING: The patient record opens. No new hospital number is created.

COMMON ERRORS: Two people with similar names; phone not stored so phone search finds nothing.

WHAT TO DO IF SOMETHING FAILS: Search again by hospital number. If you still cannot find them, do not register until you have checked with a supervisor.

SECURITY NOTES: Never open a record that is not this patient.

Registering a new patient

PURPOSE: Create a hospital record and assign a hospital number.

WHO DOES IT: Reception.

STEP-BY-STEP

1. Search first.
2. Open **Register Patient**.
3. Choose **Individual Card** or **Family Card**. The fee is billed after the visit starts. Reception does not collect payment.
4. Enter first name, last name, sex, date of birth or age, and contact and address as required.
5. Watch **Existing patients** after you type three or more characters of the first and last name. Open an existing record if it is the same person.
6. Select **Save**.

You do not type the hospital number. It is assigned when the patient is saved.

WHAT HAPPENS AFTER SAVING: A hospital number is assigned (for example an ABC number). You can start a visit and create the card fee.

COMMON ERRORS: *Similar patients found* — the system found possible matches.

WHAT TO DO IF SOMETHING FAILS: Open the existing record unless you are sure this is a new person. Use **Register anyway** only when a supervisor agrees it is a different person.

SECURITY NOTES: Do not invent a hospital number. Do not register a duplicate to “make a new visit.”

Starting a visit

PURPOSE: Record why the patient is here today.

WHO DOES IT: Reception.

STEP-BY-STEP

1. Open the patient.
2. Select **Start visit**.
3. Choose one:

What you select	Meaning
General Outpatient	Regular outpatient consultation
Follow-up	Returning for previous treatment or condition
Emergency	Urgent care
Admission	Patient is being admitted — select the **ward** shown on the screen

1. Confirm.

WHAT HAPPENS AFTER SAVING: The visit is created. You may see **Visit successfully created** and a visit number. Today's Visits lists visits that **started today**.

Outpatient and follow-up visits started today appear on the doctor’s **Today's Patients**. Emergency and admission visits do not appear on that outpatient list.

COMMON ERRORS: *This patient already has an open visit.* Start another only if they are arriving for a different kind of care.

WHAT TO DO IF SOMETHING FAILS: Check Today's Visits and the patient screen. Do not create extra visits to “reset” the list.

SECURITY NOTES: Do not start a visit on the wrong patient.

Registration card fee

PURPOSE: Send the Individual Card or Family Card bill to Accountant.

WHO DOES IT: Reception, after a visit exists.

STEP-BY-STEP

1. On the patient screen, open **Registration card fee**.
2. Confirm Individual Card or Family Card.
3. Select **Create registration fee** and confirm. Copy: *This sends the bill to Accountant. Reception does not collect payment.*

WHAT HAPPENS AFTER SAVING: Status shows **Not billed yet**, **UNPAID**, or **PAID**. When paid: *Payment received* — the receipt is on Patient Account.

One fee per visit. Submitting again reuses the existing charge.

COMMON ERRORS: Trying to collect cash in Reception.

WHAT TO DO IF SOMETHING FAILS: Direct the patient to Accountant. Open **Patient Account** to see whether the bill exists.

SECURITY NOTES: Reception cannot collect payment. Do not record payment on paper as if Abikan received it.

Card pricing

PURPOSE: Set the selling price of Individual Card and Family Card for **new** fees.

WHO DOES IT: Reception (read and write). Accountant and Hospital Admin can view. SuperAdmin can also change these prices.

STEP-BY-STEP

1. Open **Pricing**.
2. Change the naira amount if required.
3. Save. Add a reason when asked.

WHAT HAPPENS AFTER SAVING: New card fees use the new price. Price history stores the old price, new price, who changed it, and when.

COMMON ERRORS: Expecting an old unpaid or paid bill to change.

WHAT TO DO IF SOMETHING FAILS: That is correct behaviour. Yesterday’s bill keeps yesterday’s amount. See Department pricing.

SECURITY NOTES: Do not change prices unless you are authorised. Changing a price does not rewrite existing bills.

Directing the patient and Patient Account

After the visit starts, send the patient to Clinical (or Emergency / the ward as started).

Patient Account is read-only: billed so far, already received, still unpaid. Tell the patient to pay at **Accountant**.

What Reception can and cannot do

Can	Cannot
Search, register, edit name and contact	Collect payment
Start the four visit types above	Create a consultation fee
Create the registration card fee	Diagnose, prescribe, order tests, or dispense
Change Individual Card and Family Card prices	Change laboratory or pharmacy prices
Open Patient Account (read only)	Edit clinical notes, vitals, or nursing records

Known Reception limits

- There is no separate waiting-room queue. **Today's Visits** are visits that started today.
- An unpaid card fee does **not** block the doctor from opening the chart.
- Phone search only works when a phone number is stored.

Doctor — Department guide

This guide describes Clinical (Doctor) as it works in this release.

Related: User manual · Nursing · Laboratory · Pharmacy

What Clinical is for

The doctor sees today’s outpatients, records the consultation, diagnoses, orders laboratory and imaging tests, prescribes, and may request admission.

Screens: Dashboard · Today's Patients · My Patients · Patients (search) · Consultation

Dashboard and Today's Patients show the same **Today's Patients** list.

Logging in and finding patients

PURPOSE: Open today’s work.

WHO DOES IT: Staff with Clinical access.

STEP-BY-STEP

1. Sign in with your hospital username. Clinical opens.
2. Use **Today's Patients** for outpatient and follow-up visits that **started today**.
3. Use **My Patients** for people you have already attended (after a saved consultation).
4. Use header search for any patient.

WHAT HAPPENS AFTER SAVING: Opening a name continues that visit’s consultation.

COMMON ERRORS: An emergency or admission visit does not appear on Today's Patients. That is by design.

WHAT TO DO IF SOMETHING FAILS: Search by hospital number. Ask Reception whether the visit was started as General Outpatient or Follow-up.

SECURITY NOTES: Confirm name and hospital number before you write.

Consultation

PURPOSE: Record this visit’s clinical work.

WHO DOES IT: Doctor.

Consultation flow: **Review** → **Vitals** → **Consultation** → **Previous visits** → **Diagnose** → **Order** → **Prescribe** → **Admission**

What is shown

Section	What you do
Review	See allergies, known diagnoses, and medications (read-only alerts)
Vitals	Record **outpatient** vitals when an open OPD visit is selected
Consultation	Chief complaint, complaint in your own words, duration, history of present illness, clinical notes
Previous visits	Visit list and previous encounter type and date. Note text is not available in this view
Diagnosis	Search and record diagnosis

Orders	Laboratory orders and results; radiology orders and reports
Prescribe	Search medicines and prescribe
Admission	**Admit Patient** — decision only. Nursing assigns the bed

Not currently available in this release on this screen:

- Examination
- Review of Systems
- Past Medical History, Medication History, Family History, Social History
- Refer / Follow-up as a disposition
- Doctor discharge decision
- **Open full chart** (available on Nursing, Laboratory, and Pharmacy patient screens)

STEP-BY-STEP

1. Confirm patient and visit.
2. Record complaint, history, and notes.
3. Record outpatient vitals if needed (open OPD visit required).
4. Record diagnosis.
5. Order laboratory or imaging tests if needed.
6. Prescribe if needed.
7. Select **Save consultation**.

WHAT HAPPENS AFTER SAVING: Notes, vitals (OPD), diagnoses, and prescriptions for this visit are stored. Payment is **not** required to save.

COMMON ERRORS: *Visit has not started* — Reception has not started a visit. *Medications unavailable* / search unavailable — try again; if it persists, contact Hospital Admin.

WHAT TO DO IF SOMETHING FAILS: Do not start a second visit yourself unless you have Reception access. Ask Reception to start the correct visit.

SECURITY NOTES: Do not save on the wrong visit. Previous-visit encounter lines do not show note text.

Outpatient vitals versus inpatient vitals

	Outpatient (this screen)	**Inpatient (ward)**
--	----------	----------
Who records	Doctor, on an open OPD visit	Nurse, during an active admission
Who reviews	Doctor	Doctor **reads only** on consultation when the patient is admitted
If you select an inpatient visit	The OPD vitals card is hidden	Inpatient vitals panel is read-only

Doctors do **not** have Nursing write permission. Do not ask a nurse to change a time. Time is recorded automatically.

Laboratory orders and results

PURPOSE: Order hospital laboratory tests and later review results.

WHO DOES IT: Doctor orders. Laboratory processes. Accountant collects the fee.

STEP-BY-STEP

1. In **Orders**, search the hospital laboratory list.
2. Add tests and submit. A laboratory fee is sent to Accounts.

3. Tell the patient that fees are paid at Accounts.
4. When Laboratory has released a result, it appears under **Laboratory results**.

WHAT HAPPENS AFTER SAVING: The order exists for Laboratory. Processing happens in laboratory processing, not on this screen. Results return here when released.

COMMON ERRORS: Result still empty — the specimen may not have been processed yet. *Laboratory orders unavailable* / *Laboratory results unavailable* — reload; if it continues, tell Hospital Admin.

WHAT TO DO IF SOMETHING FAILS: Check Laboratory. Do not invent a result. Do not enter results in Abikan.

SECURITY NOTES: Payment is not required to place the order. **Paid does not mean the result is ready.**

Radiology orders and reports

PURPOSE: Request an approved hospital examination and later read the report.

WHO DOES IT: Doctor orders. Radiology sets price and writes the report.

STEP-BY-STEP

1. In **Orders**, choose the hospital ultrasound or radiography name.
2. Submit. Imaging fees are paid at Accounts.
3. Read **Radiology reports** when Radiology has saved a report.

WHAT HAPPENS AFTER SAVING: Radiology sees the examination. Images are **not** shown in Abikan in this release.

COMMON ERRORS: *Radiology reports unavailable* or *No report recorded for this examination.*

WHAT TO DO IF SOMETHING FAILS: Ask Radiology. Do not treat a missing image viewer as a failed order.

SECURITY NOTES: You do not collect imaging payment.

Prescriptions

PURPOSE: Prescribe medication for this visit.

Lifecycle in this release:

```
Doctor prescribes
    → prescription / drug order
    → bill created for Accounts
    → Accountant collects payment
    → Pharmacy sees Ready to dispense
    → Pharmacy dispenses
    → patient receives the medicine
```

If the patient is later admitted, scheduled inpatient doses appear on Nursing **MAR**. That is **administration**, not dispensing. Pharmacy still dispenses the supply.

When-needed (PRN) medicines are not added to the scheduled medication sheet.

COMMON ERRORS: Expecting Pharmacy to dispense before payment. The system blocks unpaid dispensing.

Admission

PURPOSE: Ask Nursing to admit the patient.

WHO DOES IT: Doctor.

STEP-BY-STEP

1. Open **Admission**.
2. Select **Admit Patient**. Copy: *The doctor decides to admit. Nursing assigns the bed. This does not collect payment.*

WHAT HAPPENS AFTER SAVING: *Admission requested. Nursing will assign a bed.* You do not choose the bed.

WHAT TO DO IF SOMETHING FAILS: Check Nursing Admissions. Do not invent a bed assignment.

After admission, on consultation you can **read** nursing notes, inpatient vitals, and MAR. You cannot record them.

Consultation fee

On an active visit, the doctor can **Create consultation fee** in naira. Status **UNPAID** / **PAID**. Copy: *Please proceed to the cashier.*

You cannot collect payment. An unpaid fee does **not** block saving the consultation.

Inpatient visibility

When the patient is admitted (or after discharge, for history):

- Inpatient vitals — **read only**
- Nursing notes — **read only**
- MAR — **read only**

Nursing notes after discharge remain readable to the doctor even when Nursing can no longer add a note.

What the doctor cannot do

- Collect payment
- Assign, transfer, or discharge a bed (Nursing)
- Record inpatient vitals, nursing notes, or MAR
- Enter laboratory results
- Dispense medicines
- Change department catalogue prices
- Open Hospital Administration analytics

Nursing — Department guide

This guide describes Nursing as it works in this release. It is a safety-critical document. Do not bypass identification, times, or administration windows.

Related: Inpatient / ward · Doctor · Pharmacy

What Nursing is for

Nursing runs the ward: who is on a bed, who is waiting for a bed, notes, vitals, and scheduled medication administration.

Screens: Dashboard · Ward Board · Patients · Admissions · Transfers · Discharges · inpatient chart

Pharmacy **dispenses** medicine. Nursing **administers** scheduled inpatient doses. These are different jobs.

Logging in

PURPOSE: Open the ward.

WHO DOES IT: Staff with Nursing access.

STEP-BY-STEP

1. Sign in with your hospital username.
2. **Nursing** opens. You do not pick a ward as a workplace.

SECURITY NOTES: Use only your account. Your name is recorded on notes, vitals, and administrations.

Ward Board, Patients, Admissions

PURPOSE: See occupancy and open the right chart.

WHO DOES IT: Nursing.

Dashboard shows occupied beds, available beds, and patients waiting for a bed, plus ward overview counts.

Ward Board shows **Ward** → **Room** → **Bed** → **Patient**. Beds are **Available** or **Occupied**. Occupied beds open the chart and offer **Transfer** and **Discharge**. Available beds open **Assign Bed**.

Patients lists occupied patients and those with **Admit requested**.

Admissions lists patients the doctor has asked to admit. Nursing assigns the bed.

Transfers and **Discharges** are shortcuts for people **currently on a bed**. They are not a history list.

STEP-BY-STEP (open a chart)

1. Confirm name and hospital number.
2. Open the patient from the board or Patients.
3. Check **Ward**, **Bed**, and status (**Occupied**, **Admit requested**, or **Not admitted**).

SECURITY NOTES: Always match the person at the bedside to the hospital number on the screen.

Assign bed, transfer, discharge

PURPOSE: Place the patient, move the patient, or free the bed.

WHO DOES IT: Nursing (bed assignment rights).

Assign bed

1. Open **Admissions**, an available bed, or the patient chart.
2. Confirm **Admit requested** when the doctor has asked to admit.
3. Choose the ward and bed shown.
4. Select **Assign Bed**.

WHAT HAPPENS AFTER SAVING: The bed is occupied. You may see *Admitted to …* Accommodation billing may start. Nursing does not collect payment. Unpaid accommodation does **not** block assign.

If the bed was just taken: *This bed has just been assigned to another patient. Please choose another bed.*

You can assign from the chart in an emergency even if the doctor has not requested admit. Use that only when the hospital requires it.

Transfer

1. From the occupied bed or Transfers, select **Transfer**.
2. Choose the new ward and bed.
3. Select **Transfer Patient**.

There is no separate transfer history screen.

Discharge

1. Confirm identity.
2. Select **Discharge**. Confirm: *Discharge this patient and release their bed?*

WHAT HAPPENS AFTER SAVING: *Patient discharged. Bed released.* Historical notes, vitals, and MAR remain readable where the system allows. You cannot record new notes, vitals, or administrations after discharge.

Unpaid accommodation does **not** block discharge. Send the family to Accountant for any unpaid bills.

WHAT NOT TO DO: Do not discharge the wrong patient. Do not try to “delete” the stay.

Inpatient chart

Header shows name, hospital number, age, ward, bed, and allergy line.

Sections that **work** in this release: alerts, current condition, **Vitals**, **Medication Administration (MAR)**, **Inpatient Nursing Notes**, doctor instructions (read-only), admission and discharge actions.

Coming soon — not currently available in this release:

- Nursing tasks
- Care plan
- Intake and output

Diagnoses may show **Diagnoses unavailable**. That is a known limit. Do not invent a diagnosis.

Open full chart opens the wider clinical chart. Daily ward work stays in Abikan.

Patient Account is read-only.

Nursing Notes

PURPOSE: Record an observation during this admission.

WHO DOES IT: Nursing, while the patient has an active bed and admission.

STEP-BY-STEP

1. Open the inpatient chart.
2. Confirm identity.
3. Select **+ Nursing Note**.
4. Write the observation, care, and response.
5. Select **Save Nursing Note**.

WHAT HAPPENS AFTER SAVING: The note is stored with **your name** and the **current date and time**, recorded automatically. You cannot edit or delete it. A later note does not replace an earlier one.

WHO CAN READ: Nursing during the active admission. The doctor can read current and historical notes.

After discharge, Nursing may see **Nursing notes unavailable** on the chart. The doctor can still read historical notes.

COMMON ERRORS: Saving a note on the wrong patient. Trying to change an old note.

WHAT TO DO IF SOMETHING FAILS: Write a new note. Do not ask anyone to change the time.

SECURITY NOTES: Time and author cannot be typed by you. Do not share an account so that another nurse's name is recorded as yours.

Vitals

PURPOSE: Record inpatient observations.

WHO DOES IT: Nursing during an active admission. Doctor reviews only.

STEP-BY-STEP

1. Open the inpatient chart → **Vitals**.
2. Confirm identity.
3. Enter the measurements you took.
4. Review warnings for unusual values. Unusual values can still be saved when they are true.
5. Confirm. Copy: observations are **added** and do **not** replace earlier readings.

Measurements in this release

Measurement	Unit
Temperature	°C
Pulse	bpm
Respiratory Rate	/min
Systolic blood pressure	mmHg
Diastolic blood pressure	mmHg
SpO_2	%
Weight	kg
Height	cm

BMI may display from weight and height. It is **calculated on screen and is not saved** as a stored observation.

Both systolic and diastolic must be entered together. Systolic must be higher than diastolic. Values outside the allowed range are refused.

WHAT HAPPENS AFTER SAVING: Time is recorded automatically. Each recording is a **new** observation. History is kept. Later measurements do not overwrite earlier ones.

After discharge you can read history. You cannot record new vitals (*Vitals can be recorded during an active admission*).

OPD versus ward: The doctor may record outpatient vitals on an OPD visit. Those are not the same as ward vitals. The condition summary may show a latest reading from a wider history; the **Vitals** section shows this admission's recordings.

WHAT NOT TO DO: Do not change timestamps. Do not record another nurse's work under your name.

Medication administration (MAR)

PURPOSE: Give or document a **scheduled inpatient** dose.

WHO DOES IT: Nursing during an active admission.

```
Doctor prescribes (inpatient)
    → medication schedule is created for eligible orders
    → scheduled dose appears on MAR
    → Nurse administers (or records not given)
    → administration is stored
```

When-needed (PRN) medicines are not added to the scheduled medication sheet. They cannot be administered from MAR in this release.

Outpatient-only orders and incomplete prescriptions also do not become scheduled slots.

Window

Normal window: **30 minutes before** the scheduled time through **60 minutes after**.

- Inside the window: **Administer** for **Given** (note optional).
- Outside the window: use **Administer outside scheduled window**. A **reason is required**, including when the dose was given.
- **Not done**, **Refused**, and **Held** always need a reason.

Actual time is recorded automatically. You cannot type or change it. Scheduled time and actual time are stored separately.

Outcomes

You select	Meaning
Given	Dose administered
Not done	Dose not given
Refused	Patient refused
Held	Held

A finished slot cannot be administered again. After discharge the sheet is **historical** — readable, no new administrations.

Duplicate protection: Repeating the same save does not create a second administration.

Patient isolation: A slot for another patient is refused (*This scheduled dose is for another patient*).

COMMON ERRORS: *This dose is outside the normal administration window.* *Medication administration unavailable.* Empty MAR because the order is PRN or not inpatient.

WHAT TO DO IF SOMETHING FAILS: Confirm you are on the correct patient and admission. Use the outside-window action with a true clinical reason when the time is outside the window. Do not invent a scheduled time. Do not administer from Pharmacy.

SECURITY NOTES: Do not bypass the window. Do not administer after discharge. Do not record a dose you did not give.

What Nursing cannot do

- Collect payment
- Change accommodation tariffs or department catalogue prices
- Dispense from Pharmacy
- Write diagnoses or prescriptions
- Request admission (doctor) or process laboratory specimens
- Edit or delete saved notes, vitals, or MAR rows
- Administer PRN as scheduled MAR
- Backdate times

Known Nursing limits

- Diagnoses may show unavailable
- Tasks, care plan, and intake/output show **Coming soon**
- No transfer or discharge history list
- Ward and room may show the same name in assign/transfer
- Wards without an accommodation tariff show **0** until a price is set

Inpatient and ward operations

This guide is the shared ward picture. Detailed Nursing steps are in Nursing. Doctor steps are in Doctor.

Who does what

Role	What they do on an admission
Doctor	Clinical decisions, orders, prescriptions, **Admit Patient**
Nurse	Assign / transfer / discharge bed, notes, vitals, MAR
Reception	Registration and starting an **Admission** visit when that is the front-desk path
Accountant	Collect accommodation and other bills
Pharmacy	Dispense paid medicines (not MAR)
Laboratory	Specimen and results in laboratory processing
Radiology	Price and report (images not in Abikan)
Hospital Admin	Staff and counts — not bedside care

Do not assign a job the application does not support.

Admission

```
Doctor: Admit Patient
    → Admission requested
    → Nursing: Assign Bed
    → Occupied
```

Reception may also start an **Admission** visit and choose a ward. That does not replace Nursing bed assignment.

The doctor does not choose the bed and does not collect payment.

Ward and bed

Nursing **Ward Board** shows ward, room, and bed as **Available** or **Occupied**.

Transfer moves an occupied patient to another bed. There is no transfer history list.

Discharge frees the bed. Nursing performs discharge. Doctor discharge decision is **not currently available in this release**.

Unpaid accommodation does not block assign, transfer, or discharge.

Inpatient chart

Nursing opens the inpatient chart for notes, vitals, and MAR.

The doctor opens the same patient in Clinical and **reads** notes, vitals, and MAR. The doctor does not write those.

After discharge, history remains. Nursing cannot add new notes, vitals, or administrations.

Vitals and MAR

See Nursing.

- Time is automatic.
- Each vitals recording is new; history is kept.
- MAR is **scheduled inpatient** doses only. PRN is not on the sheet.
- Pharmacy dispense and MAR administer are different.

Laboratory, pharmacy, accommodation

Laboratory and imaging orders still come from the doctor. Processing and reporting stay in those departments.

Pharmacy dispenses paid prescriptions. The ward does not dispense from MAR.

Accommodation: SuperAdmin sets tariff. Charge may follow assignment. Old dollar bills stay in dollars. Tariff **0** means the ward is not configured yet.

Discharge

```
Nursing: confirm identity
    → Discharge
    → Bed released
    → Accountant: collect remaining bills
    → Records remain
```

Do not discharge a patient from another environment's important fixtures during training. On UAT, discharge only the synthetic admission created for the test.

Laboratory — Department guide

This guide describes Laboratory as it works in this release.

Abikan does not perform laboratory processing. Specimen collection, accession, testing, result entry, and validation happen in **laboratory processing** (the hospital laboratory system). Abikan shows orders, payment state, and released results.

Related: Doctor · Billing · OpenELIS administration notes (technical)

What Laboratory is for in Abikan

Screens: Dashboard · Pending Orders · Processing · Completed Results · Search Patient · Test Pricing

In Abikan	In laboratory processing
See which tests the doctor ordered	Collect the specimen
See unpaid / paid	Accession and process the sample
Open **Open laboratory processing**	Enter and validate the result
See a result after it is released	Source of the result
Set hospital test selling prices	Catalogue of processing methods

Abikan **does not write** processing records. Do not try to type a result in Abikan.

Logging in

Sign in with your **Abikan** laboratory username. That opens the Laboratory board.

Laboratory processing uses a **separate sign-in**. Your Abikan account does not log you into processing. Hospital-specific processing accounts and roles are set by the laboratory administrator. See hospital OpenELIS training — this manual does not invent those clicks.

How an order appears

```
Doctor creates a laboratory order on the visit
    → Laboratory sees the order on the board
    → Patient pays at Accountant (when billed)
    → Laboratory collects and processes in laboratory processing
    → Validated result returns
    → Doctor and Laboratory see the result in Abikan
```

Payment is **not** required for the doctor to place the order. **Paid does not mean the sample was collected. Paid does not mean the result is ready.**

Pending, Processing, Completed

PURPOSE: Know which orders need work and which have results.

WHO DOES IT: Laboratory.

STEP-BY-STEP

1. Open **Pending Orders** for new work.

2. Open **Processing** for work already in progress.
3. Open **Completed Results** for released results.
4. Open an order. Confirm **patient, hospital number, visit, and test**.
5. Select **Open laboratory processing** to collect, accession, test, enter, and validate — using your processing account and hospital procedure.

The board groups orders by processing status. Unrecognised states are shown honestly, not invented.

The board can take **several seconds** to load. That is a known limit.

WHAT HAPPENS AFTER SAVING (in processing): After the result is validated and released, it appears in Abikan for Laboratory and the doctor. Abikan does not store a second result file.

COMMON ERRORS: Order not on the board — the doctor may not have submitted, or you opened a different patient. Result empty — processing not finished. *Orders unavailable* / *Results unavailable* — reload; if it continues, tell Hospital Admin.

WHAT TO DO IF SOMETHING FAILS: Confirm the hospital number with the doctor. Do not fabricate a result. Do not skip validation.

SECURITY NOTES: Never enter another patient's result. Never use a shared processing login for clinical work.

Specimen collection, accession, testing, validation

These steps are **not performed in Abikan**.

Follow your hospital's laboratory processing training for:

- Samples to collect
- Collect sample
- Accession
- Results entry
- Validation / completion

This release does not document those screens as Abikan procedures. If a new test was added to the hospital list, the laboratory administrator may still need to finish specimen and range setup in processing, or samples can sit untested.

Hospital UAT: process a **new** result on this hospital stack. Do not treat an old result from another environment as proof.

Test pricing

PURPOSE: Set the selling price of hospital laboratory tests for **new** fees.

WHO DOES IT: Laboratory (read and write). Accountant and Hospital Admin can view. SuperAdmin can also change these prices.

STEP-BY-STEP

1. Open **Test Pricing**.
2. Find the hospital test (this is the ordering list, not the full processing dictionary).
3. Change the naira price and save.

WHAT HAPPENS AFTER SAVING: New charges use the new price. Bills already created do **not** change.

Four tests are **not currently available to order** from Abikan: Inorganic Phosphate, Testosterone, Progesterone, Estradiol.

Some tests are marked **Outsourced**. There is no separate outsourced workflow in Abikan.

What Laboratory cannot do

- Collect payment (*Please proceed to the cashier. Laboratory does not collect money.*)
- Enter or validate results in Abikan
- Collect specimens in Abikan
- Change Reception or Pharmacy prices
- Open Administration analytics

Known Laboratory limits

- Fresh result should be proven during hospital UAT
- Board can be slow
- Abikan does not write to laboratory processing
- Paid ≠ collected ≠ resulted

Pharmacy — Department guide

This guide describes Pharmacy as it works in this release.

Pharmacy **dispenses** a paid prescription. Nursing **administers** a scheduled inpatient dose on MAR. Do not mix these jobs.

Related: Doctor · Accountant · Nursing

What Pharmacy is for

Screens: Dashboard · Prescriptions · Ready to Dispense · Dispensed · Drug Catalogue · Drug Pricing · Inventory

```
Doctor prescribes
    → bill is created
    → Accountant collects payment
    → Ready to dispense
    → Pharmacy dispenses
    → stock is updated
```

Logging in and finding work

Sign in with your Pharmacy username.

Dashboard puts **Ready to dispense** first.

Prescriptions lists prescription work. **Ready to Dispense** is paid and waiting. **Dispensed** is already given out.

Confirm **patient, hospital number, and medicine** before you dispense.

Payment state

What you see	Meaning
Unpaid / payment required	Do **not** dispense. Send the patient to Accountant
Partial payment	**Not** ready to dispense in this release
Ready to dispense	Posted bill is fully paid; stock movement not yet completed
Dispensed	You have completed dispense; stock has moved

Pharmacy has **no payment button**.

Dispensing

PURPOSE: Give the patient the prescribed medicine and update stock.

WHO DOES IT: Pharmacy.

STEP-BY-STEP

1. Open the prescription. Confirm identity and medicine.
2. Check payment is **Ready to dispense**.
3. Check available stock.
4. Select **Dispense**. Confirm: *This will mark the prescription as dispensed and update pharmacy stock.*

This release dispenses the **full** quantity. Partial dispense is not offered.

WHAT HAPPENS AFTER SAVING: Status becomes **Dispensed**. Repeating the action does not dispense twice (*This medication has already been dispensed*).

COMMON ERRORS

- *Payment required before dispensing. Please proceed to the cashier.*
- *This medication has already been dispensed.*
- *Not enough stock to dispense this medication.*

WHAT TO DO IF SOMETHING FAILS: Do not override payment. Do not dispense unpaid medication. Do not invent stock. If stock is short, use Inventory (if you have that right) or tell Hospital Admin.

SECURITY NOTES: Do not share accounts. Do not dispense on the wrong hospital number.

Drug Catalogue and Drug Pricing

Drug Catalogue adds and edits hospital medicines, including selling price.

Drug Pricing changes selling prices for **new** sales. Staff see **₦**.

Changing today's price does **not** change a bill already created.

Pharmacy can write pharmacy prices. Reception and Laboratory cannot. Accountant and Hospital Admin can view. SuperAdmin can also change these prices.

Inventory

Inventory shows stock. **Receive stock** and **Adjust stock** require the hospital inventory right. Adjustment reasons include incorrect opening stock, physical count, damaged, expired, lost/missing, and correction.

Purchase orders and supplier payment are **not currently available in this release**.

Prescription versus MAR

Pharmacy	Nursing MAR
Dispense the supply after **payment**	Administer a **scheduled inpatient** dose
Updates stock	Records Given / Not done / Refused / Held
Counter	Ward

Do not present MAR rows as pharmacy dispensing.

What Pharmacy cannot do

- Collect payment
- Administer MAR
- Enter laboratory results
- Change Reception or Laboratory prices
- Dispense unpaid or partially paid prescriptions

Accountant — Department guide

This guide describes the cashier as it works in this release.

Related: Billing and payments · Reception · Pharmacy

What Accountant is for

Accountant collects money on posted bills and opens receipts.

Screens: Cashier · Patient Accounts (search) · one-patient collect screen · Patient Account (read-only money view)

```
Department creates a charge
    → bill is posted
    → Accountant collects payment
    → receipt
    → department shows Paid (Pharmacy shows Ready to dispense)
```

Logging in

Sign in with an account that has Accountant access. **Cashier** opens: *Search a patient to collect payment.*

Finding the bill and collecting

PURPOSE: Record what was collected.

WHO DOES IT: Accountant.

STEP-BY-STEP

1. Search by name, hospital number, or phone.
2. Confirm identity.
3. Review bills that are still unpaid.
4. Collect the amount received.
5. Confirm. Open or print the receipt when offered.

WHAT HAPPENS AFTER SAVING: The bill shows paid (or remaining unpaid if only part was collected, where partial payment is offered). Pharmacy can then dispense a **fully** paid prescription. Laboratory and Radiology still must finish their own work.

COMMON ERRORS: Patient not found — try hospital number. Nothing to collect — the department may not have created the charge, or the bill is already paid. Mixed currencies — *Select invoices in the same currency.*

WHAT TO DO IF SOMETHING FAILS: Ask the originating department whether they created the fee. Do not mark a bill paid on paper only. Do not ask Pharmacy to dispense unpaid medication.

SECURITY NOTES: Do not share the cashier account. Do not skip identity.

Billed, collected, outstanding

These words mean **money on bills**, not hospital “revenue.”

Word	Meaning in Abikan
Billed / Amount	Amount on bills created so far

Collected / Paid	Amount already received
Outstanding / Still unpaid	Amount still unpaid **now**

On **Patient Account**:

- **Amount** — *Billed so far*
- **Paid** — *Already received*
- **Outstanding** — *Still unpaid*

On **Cashier**: **Collected today** and **Still unpaid**.

Hospital Admin **Outstanding** is a **current snapshot**, not the balance as of a past date.

Which bills you collect

Posted bills from:

- Registration cards (Individual Card, Family Card)
- Consultation fees
- Laboratory fees
- Radiology examination prices
- Pharmacy prescriptions
- Accommodation (daily ward/room charges)

Draft bills that are not posted are not collectable.

Refunds are not currently available in this release.

Accommodation

Accommodation tariffs are set by **SuperAdmin** under Administration → **Accommodation Prices** (naira daily price per ward/room).

A charge can be created when Nursing assigns a bed. Changing today's tariff does **not** rewrite a bill already posted.

New accommodation bills use the naira accommodation list. **Older accommodation bills that were in dollars stay in dollars.**

Wards without a configured tariff may show **0** until a price is set.

Unpaid accommodation does **not** block Nursing assign, transfer, or discharge. Still collect what is owed.

Prices you cannot change

Accountant can **view** Reception, Laboratory, and Pharmacy prices. Accountant **cannot** change them.

Changing a catalogue price never rewrites an old bill. If a patient disputes an old amount, look at the bill from the day it was created.

What Accountant cannot do

- Dispense medicines
- Record clinical notes, vitals, or MAR
- Enter laboratory or radiology results
- Create registration, laboratory, or consultation charges from the cashier screen

- Change department catalogue prices
- Process refunds in this release

Radiology — Department guide

This guide describes Radiology as it works in this release.

Images are not shown in Abikan. Do not invent PACS or viewer steps.

Related: Doctor · Billing · Radiology workflow notes

What Radiology is for in Abikan

Screens: Radiology (dashboard) · Worklist · patient imaging · examination detail

You can	You cannot in this release
See pending examinations (order without a report)	View or upload scan images in Abikan
Set a naira price for that examination	Collect payment
Write or edit a free-text report	Cancel or delete the order from Radiology
See Price pending / Unpaid / Paid	Treat Paid as “scan done”

The doctor orders from a hospital-approved ultrasound and radiography list (42 services). ECG, CT, MRI, and other names not on that list are **not currently available** on the Radiology menu.

Worklist and report

PURPOSE: Price the examination and record the report.

WHO DOES IT: Radiology.

STEP-BY-STEP

1. Open **Worklist**. Empty state: *No pending Radiology examinations.*
2. Open the examination. Confirm patient, hospital number, visit, and examination name.
3. **Set Price** / **Save Price** in naira. *Accounts collects payment. You can still write the report if unpaid.*
4. **Write Report** or **Edit Report**. Free text. Save.

WHAT HAPPENS AFTER SAVING: The doctor can read the report on that visit and examination. Payment status updates after Accountant collects.

COMMON ERRORS: *Order unavailable.* Report not visible to the doctor — confirm you saved on the correct examination.

WHAT TO DO IF SOMETHING FAILS: Do not attach the report to a different order. Do not tell staff to open a hidden imaging archive as if it were Abikan.

SECURITY NOTES: You do not collect payment. Do not share accounts.

Imaging archive

The hospital may run a separate imaging archive. In this release Abikan does **not** show studies, does not replace the archive screens, and does not require a scan to be paid before you can write a report.

PAID does not mean the scan was performed.

If a modality and archive are connected later, that is hospital imaging operations — **not currently available in this release** as an Abikan workflow.

What Radiology cannot do

- Collect payment
- Cancel the order from this desk
- Display images
- Change Reception, Laboratory, or Pharmacy catalogue prices

Hospital Admin — Department guide

This guide describes Hospital Administration as it works in this release.

Related: Roles · Hospital overview · Deployment

What Hospital Admin is for

Screens: Dashboard · Analytics (Hospital overview) · Audit log (Staff activity) · Staff accounts · Department access · Accommodation Prices (SuperAdmin write) · System Settings (**Coming soon**)

Hospital Admin manages **staff and hospital counts**. It does not replace Reception, Clinical, Nursing, Laboratory, Pharmacy, Radiology, or Accountant.

Who can open Administration

Account	Access
Hospital Administrator	Home, staff accounts, department access, hospital overview, staff activity
SuperAdmin	All of the above, every department, and **Accommodation Prices** write
Doctor, Nurse, Laboratory, Pharmacy, Accountant	Denied unless they also have admin rights

Frontend hiding is not enough. The server also requires the matching administration rights.

Staff accounts

PURPOSE: Create and disable hospital users.

WHO DOES IT: Hospital Admin. Only SuperAdmin can create another Hospital Administrator.

STEP-BY-STEP

1. Open **Staff accounts**.
2. **Add staff**: full name, username, initial sign-in password (at least 8 characters), and account type.
3. Assign departments with the checkboxes. Do not invent extra technical rights here.
4. Save. The initial password is **not shown again** and is not stored on Staff activity.

Clinical profile: some departments need a clinical profile (**Ready**, **Clinical profile needed**, or **Not required**). The system links it when the department requires it.

You can disable or enable an account. A disabled account cannot sign in.

SECURITY NOTES: Give each person their own account. Do not reuse development passwords. Do not write passwords into email or this manual. Tell the staff member to change the initial password according to hospital policy.

Department access

Approved access only. Departments and roles are the hospital options on the screen. Technical permissions cannot be assigned here.

Hospital Administrator rows cannot have department access changed here — that is technical support.

Hospital overview (Analytics)

PURPOSE: See hospital-wide counts.

WHO DOES IT: Hospital Admin.

STEP-BY-STEP

1. Open **Analytics**.
2. Choose **Today**, **Yesterday**, **This week**, **Last week**, **This month**, **Last month**, or a custom range. Dates use **Ilorin time**.

Label	Date filter?
Patients registered	Yes
Outpatient visits (Reception subsection)	Yes
Consultations	Yes
Admissions	Yes
Discharges	Yes
Laboratory orders	Yes
Prescriptions	Yes
Dispensed	Yes
Billed	Yes (~~₦~~)
Collected	Yes (~~₦~~)
Current occupancy	**No — now**
Current occupancy rate	**No — now**
Outstanding	**No — now**

Current occupancy and **Current outstanding** are **snapshots of now**. They do not mean occupancy or unpaid amounts as of the selected dates.

If a figure cannot be read, the screen shows **Unavailable**, not a fake zero.

These counts are **not** “revenue.” Billed, collected, and outstanding are bill totals.

Not shown: named patients, new versus returning split, aged receivables, historical occupancy.

Staff activity

PURPOSE: See who changed staff accounts and access.

Recorded: created staff / hospital administrator, disabled or enabled account, updated / assigned / removed department access, linked clinical profile.

Not recorded: clinical chart views, patient care, laboratory processing, payments.

Passwords, cookies, and secrets are **not displayed**.

Accommodation Prices

SuperAdmin sets the daily naira tariff per ward/room.

Hospital Admin may see the menu item and be **denied** write. That is expected.

Wards without a tariff show **0** until configured. Changing a tariff does not rewrite old bills.

System Settings

Not currently available in this release (Coming soon).

What Hospital Admin cannot do (unless also given that department)

- Collect payment
- Record clinical or nursing data
- Process specimens or dispense
- Change Reception / Laboratory / Pharmacy catalogue prices (view only)

SuperAdmin **can** change those catalogue prices and accommodation tariffs. That is an accepted limitation of this release. Use it carefully.

Clinical and hospital workflows

These diagrams describe **this release only**. Steps that are not implemented are marked.

For staff how-to, use the department manuals. For money, use BILLING_AND_PAYMENTS.md.

---

1. New patient

```
Reception: Search first
    ↓ (no safe match)
Reception: Register Patient
    → choose Individual Card or Family Card
    → enter name, sex, birth date or age, contact, address
    → Save
    → hospital number is assigned
    ↓
Reception: Start visit
    ↓
Reception: Create registration fee (after a visit exists)
    ↓
Accountant: Collect payment when the patient is ready
    ↓
Direct the patient to Clinical (or Emergency / ward as started)
```

If similar patients appear, open the existing record unless you are sure this is a new person. See Reception.

---

2. Returning patient

```
Reception: Search by name, hospital number, or phone
    ↓
Open the existing patient
    ↓
Start a new visit only if they need a different kind of care
    ↓
Card fee is per visit when Reception creates it
    ↓
Clinical: Today's Patients (outpatient visits started today)
    or search if the visit type is Emergency or Admission
```

---

3. OPD consultation

```
Reception starts General Outpatient or Follow-up
    ↓
Doctor: Today's Patients → open the name
    ↓
Review allergies / known problems
    ↓
Record OPD vitals on this open outpatient visit (if needed)
    ↓
Chief complaint, history of present illness, clinical notes
    ↓
Diagnosis (optional)
    ↓
Laboratory and / or radiology orders (optional)
    ↓
Prescription (optional)
    ↓
Admit Patient (optional)
    ↓
Save consultation
    ↓
Create consultation fee (optional) → Accountant collects
```

Not currently available on this screen: Examination, Review of Systems, Past Medical History, Medication History, Family History, Social History, Refer, doctor follow-up disposition, doctor discharge decision.

---

4. Laboratory

```
Doctor: Order → Laboratory → submit tests
    → laboratory fee is sent to Accounts
    ↓
Laboratory: Pending Orders (and patient search)
    ↓
Laboratory staff: Open laboratory processing
    → collect specimen, accession, test, enter result, validate
    (those steps are not done inside Abikan)
    ↓
Result returns to Abikan
    ↓
Doctor: Laboratory results on the same visit
```

Abikan does **not** collect the specimen or enter the result. See Laboratory.

A new result should be processed during hospital UAT.

---

5. Pharmacy

```
Doctor: Prescribe
    ↓
Charge / invoice is created for the prescription
    ↓
Accountant: Collect payment (full payment)
    ↓
Pharmacy: Ready to Dispense
    ↓
Pharmacist: Confirm identity → Dispense
    ↓
Stock moves; status becomes Dispensed
    ↓
Patient receives the medicine at the pharmacy
```

Unpaid and partly paid prescriptions are **not** ready to dispense. This is not the same as Nursing MAR.

---

6. Admission

```
Doctor: Admit Patient
    → "Admission requested. Nursing will assign a bed."
    ↓
Nursing: Admissions (Admit requested)
    ↓
Nursing: Assign Bed
    → inpatient stay starts
    → accommodation charge may be created
    ↓
Accountant: Collect accommodation when billed
```

Reception can also start an **Admission** visit and choose a ward. That is a Reception visit start, not the Doctor admit button.

Doctor does **not** choose the bed. Nursing does **not** decide the clinical admission.

---

7. Nursing (inpatient chart)

```
Ward Board or Patients → open the patient
    ↓
Confirm name and hospital number
    ↓
Implemented:
    Nursing Notes
    Vitals
    Medication Administration (scheduled doses)
    Assign / Transfer / Discharge
    ↓
Coming soon (not stored):
    Nursing tasks
    Care plan
    Intake and output
```

---

8. Vitals

Outpatient (Doctor)

```
Open OPD visit
    ↓
Consultation → Vitals
    ↓
Enter measurements → Save consultation
```

Inpatient (Nursing)

```
Admitted patient with a bed
    ↓
Inpatient chart → Vitals → Record vitals
    ↓
Review and record
    → time and nurse are stored automatically
    → a new reading is added; earlier readings stay
```

Doctor can **read** inpatient vitals. Doctor cannot record inpatient vitals.

---

9. MAR (scheduled medication)

```
Doctor prescribes an inpatient scheduled medicine
    ↓
Schedule / dose slots are prepared for the admission
    ↓
Nurse: Medication Administration
    ↓
Within the normal window: Administer (Given / Not done / Refused / Held)
    ↓
Outside the window: Administer outside scheduled window + reason
    ↓
Actual time is recorded automatically
```

When-needed (PRN) medicines are **not** added to the scheduled sheet. Pharmacy dispense is a different workflow.

---

10. Radiology

```
Doctor: Order → Radiology → submit examination
    ↓
Radiology: Worklist
    ↓
Set price in naira (Accounts collects)
    ↓
Write or edit the report (free text)
    ↓
Doctor: Radiology reports on the visit
```

Images are not currently available in Abikan. Connecting a scanner and viewing files needs hospital imaging setup outside this manual. Do not invent a PACS procedure from Abikan.

---

11. Discharge

```
Nursing: Ward Board (or Discharges list) → Discharge
    → confirm
    → bed is released
    ↓
Historical notes, vitals, and MAR remain readable
    (Nursing cannot add new notes / vitals / administrations after discharge)
    ↓
Doctor can still read historical inpatient records on the consultation
```

Doctor discharge decision is **not currently available**. Unpaid accommodation does **not** block discharge.

---

12. Billing and payment

```
Department creates a charge
    ↓
Bill appears for Accountant
    ↓
Accountant: Collect payment → receipt
    ↓
Department screen shows Paid
    (Pharmacy: Ready to Dispense, then Dispense)
```

See BILLING_AND_PAYMENTS.md.

Billing and payments

Staff language. The billing engine stores the commercial record. Abikan is the cashier and department charge screens.

Related: Accountant · Department pricing

How a bill is born

```
Catalogue selling price (today)
    → department creates a charge
    → posted bill
    → Accountant collects payment
    → collected
```

Step	Meaning
Catalogue price	Current selling price on the hospital item
Charge	That price copied onto this visit / order / day
Bill	Posted amount the cashier can collect
Payment	Money recorded as received
Collection	Same as payment from the cashier's point of view

Abikan does not keep a second ledger.

Price snapshot

Changing today's catalogue price does not change yesterday's bill.

The amount is copied when the charge is created. Later price edits leave that bill, its payments, and its outstanding amount unchanged.

Price history (who changed a catalogue price) is separate from the bill.

Billed, collected, outstanding

Do **not** call these "revenue." They are bill totals.

Term	Meaning
Billed	Amount on bills created
Collected	Amount received
Outstanding	Amount still unpaid **now**

Hospital overview **Outstanding** and **Current occupancy** are snapshots of now, not historical values for the selected date range.

Who creates the charge

Charge	Who creates it	When
Individual Card / Family Card	Reception	After a visit exists

Consultation fee	Doctor	On the visit
Laboratory fee	With the laboratory order	When the doctor submits
Radiology price	Radiology	Per examination
Pharmacy	From the prescription	Commercial record after prescribe
Accommodation	After ward assignment	Daily room charge

Only Accountant **collects**.

Paid does not mean the consultation is finished, the specimen is collected, the medicine is dispensed, or the scan was performed.

Pharmacy dispenses only when the prescription bill is **fully** paid. Partial payment is not ready to dispense.

Currency

- Staff pricing screens and new departmental bills use **naira (₦)**.
- The financial **company currency may still be US dollars**.
- **Historical accommodation invoices in dollars stay in dollars.** New accommodation uses the naira accommodation list.
- If Patient Account mixes currencies, collect same-currency bills together.

Accommodation

SuperAdmin sets daily naira tariffs. Unconfigured wards may show **0**. Unpaid accommodation does not block Nursing assign, transfer, or discharge.

Refunds

Not currently available in this release.

User roles and permissions

This matrix is from **server rules** in this release, not from a hidden menu. If the screen hides a button but the server allows it, the server wins. If the screen shows a link but the server refuses, the server wins.

Related: Hospital Admin · Department pricing

Unauthenticated requests: **Sign in required.** Everything below is **NO**.

How access is granted

Hospital Admin assigns **departments** on Staff accounts. That grants the hospital role for that desk.

SuperAdmin (technical) may open every department and is treated as a departmental actor on many server paths (including catalogue price writes).

Hospital Admin opens Administration only. They cannot write department catalogue prices. They cannot write accommodation tariffs.

Frontend Pharmacy and Radiology switches for SuperAdmin are offered on the department switcher; the server also accepts SuperAdmin on those APIs.

Capability key

Value	Meaning
YES	Allowed on the server for that job
NO	Refused
READ ONLY	Can view, cannot change
OWN DEPT	Only that department's list
N/A	Not a function of this release

DELETE/VOID of clinical notes, vitals, and MAR is **NO** for everyone (immutable). Radiology cannot cancel an order from the Radiology desk.

APPROVE of laboratory results is **not in Abikan**. Validation is laboratory processing.

Matrix

Capability	SuperAdmin	Hospital Admin	Reception	Doctor	Nurse	Laboratory	Pharmacy	Accountant	Radiology
Open own department	YES	YES (Admin)	YES	YES	YES	YES	YES	YES	YES
Open other operational departments	YES	NO	NO	NO	NO	NO	NO	NO	NO
Register / edit demographics	YES	NO	YES	NO	NO	NO	NO	NO	NO
Start visit	YES	NO	YES	NO	NO	NO	NO	NO	NO

Create registration card charge	YES	NO	YES	NO	NO	NO	NO	NO	NO
Consultation write (notes, OPD vitals, diagnosis, Rx, lab/rad order)	YES	NO	NO	YES	NO	NO	NO	NO	NO
Admit decision	YES	NO	NO	YES	NO	NO	NO	NO	NO
Assign / transfer / discharge bed	YES	NO	NO	NO	YES	NO	NO	NO	NO
Nursing notes / inpatient vitals / MAR **write**	YES	NO	NO	NO	YES	NO	NO	NO	NO
Nursing notes / inpatient vitals / MAR **read**	YES	NO	NO	YES	YES*	NO	NO	NO	NO
Laboratory board	YES	NO	NO	NO	NO	YES	NO	NO	NO
Enter lab result in Abikan	NO	NO	NO	NO	NO	NO	NO	NO	NO
Dispense	YES	NO	NO	NO	NO	NO	YES	NO	NO
Collect payment	YES	NO	NO	NO	NO	NO	NO	YES	NO
Radiology price / report	YES	NO	NO	NO	NO	NO	NO	NO	YES
Reception card prices	YES	READ ONLY	YES	NO	NO	NO	NO	READ ONLY	NO
Laboratory test prices	YES	READ ONLY	NO	NO	NO	YES	NO	READ ONLY	NO
Pharmacy drug prices	YES	READ ONLY	NO	NO	NO	NO	YES	READ ONLY	NO
Accommodation tariffs	YES	NO	NO	NO	NO	NO	NO	NO	NO
Hospital overview	YES	YES	NO	NO	NO	NO	NO	NO	NO
Staff activity	YES	YES	NO	NO	NO	NO	NO	NO	NO
Staff accounts / department access	YES	YES	NO	NO	NO	NO	NO	NO	NO
Create Hospital Administrator	YES	NO	NO	NO	NO	NO	NO	NO	NO
Patient Account (money view)	YES	YES†	YES	YES	YES	YES	YES	YES	YES
Delete / void clinical records	NO	NO	NO	NO	NO	NO	NO	NO	NO
Refund	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

* Nurse reads notes on the **active** admission. After discharge, Nursing may be unable to load notes; the doctor can still read history.

† Hospital Admin can open Patient Account if they also open a patient workspace; they do not collect payment.

Pharmacy **Receive / Adjust stock** needs the extra inventory right. Without it: view and dispense only.

Unauthenticated

VIEW, CREATE, EDIT, DELETE, APPROVE, DISPENSE, ADMINISTER, PRICE MANAGEMENT, ANALYTICS, AUDIT, STAFF ADMIN: **NO**.

Troubleshooting

For hospital staff. Do not change hidden records or share accounts.

I cannot log in

Possible cause: Wrong username or password; account disabled; network.

Check: Caps lock. Your own username. Hospital address.

Do: Try again. Ask Hospital Admin whether the account is disabled.

Contact Hospital Admin: After two or three failures, or if many staff fail at once.

Do not: Use another person’s account or a development password.

The EMR is unavailable / Unable to connect

Possible cause: Network, power, or the hospital system is down.

Check: Another computer on the hospital network. Whether others are affected.

Do: Wait and try again. If everyone is down, contact Hospital Admin.

Do not: Recreate patients on paper without a supervisor’s downtime plan.

I cannot find the patient

Possible cause: Spelling; they were never registered; you searched phone but no phone is stored.

Check: Hospital number. Alternate spelling. Ask Reception.

Do: Search again. Register only after a supervisor agrees this is a new person.

Do not: Create a duplicate record to “get on with it.”

The patient is not appearing in my department

Possible cause: Wrong visit type; not admitted; not ordered; not paid (Pharmacy).

Check:

- **Doctor Today's Patients:** visit must be General Outpatient or Follow-up **started today**. Emergency and admission are excluded.
- **Nursing:** doctor admit and/or bed assignment.
- **Laboratory:** doctor submitted the order.
- **Pharmacy Ready to dispense:** bill must be **fully** paid.
- **Radiology:** an imaging order exists.

Do: Ask the department that starts that step.

Do not: Start extra visits or extra registrations to force the list.

The medication is not appearing on MAR

Possible cause: When-needed (PRN) medicine; outpatient order; not admitted; schedule not created yet.

Check: The patient is occupied on a bed. The prescription is an inpatient scheduled order, not PRN.

Do: PRN is **not** on the scheduled sheet in this release. Use the doctor and hospital policy outside MAR. Reload the chart.

Do not: Administer from Pharmacy as if it were MAR.

I cannot administer the medication

Possible cause: Outside the window; already recorded; discharged; wrong patient.

Check: Scheduled versus actual window (30 minutes early / 60 minutes late). Slot status. Admission still active.

Do: Use **Administer outside scheduled window** with a true reason when the time is outside the window. Confirm identity.

Do not: Change the clock or ask anyone to backdate.

The medication is outside the administration window

Possible cause: You are more than 30 minutes early or more than 60 minutes late.

Do: Use the outside-window action and record the clinical reason. Time is stored automatically.

Do not: Skip the reason or record a false time.

The laboratory order is not visible

Possible cause: Order not submitted; different patient; board still loading.

Check: Doctor consultation orders. Hospital number. Wait several seconds and refresh — the board can be slow.

Do: Ask the doctor to confirm submit.

Do not: Create a new patient.

The laboratory result is not visible

Possible cause: Specimen not collected; result not validated; processing not finished.

Check: Laboratory processing, not only Abikan.

Do: Complete collection, testing, and validation in laboratory processing. Process a **new** result during UAT.

Do not: Type a result into Abikan or invent a value.

The prescription is not available for dispensing

Possible cause: Unpaid or partial payment; already dispensed; no stock.

Check: Accountant. Stock on the prescription.

Do: Send the patient to pay. Do not dispense unpaid medication.

Do not: Bypass the block.

The patient has not paid

Possible cause: Charge not created; cashier not collected.

Check: Patient Account. Which department should have created the fee.

Do: Reception / Doctor / Radiology / Pharmacy path as appropriate, then Accountant.

Do not: Treat unpaid as dispensed or as a completed laboratory result.

The accommodation charge looks wrong

Possible cause: Tariff was **0**; old bill in dollars; price changed after the bill.

Check: Whether the bill is old. Whether that ward has a tariff.

Do: Collect the billed amount. Ask SuperAdmin to set missing tariffs for **new** days. Do not expect an old bill to change.

Do not: Edit hidden financial records.

The price I changed did not change an old bill

Possible cause: This is how the system works.

Do: Explain that the bill keeps the price from the day it was created.

Do not: Create a fake second bill to “fix” history unless a supervisor starts a formal correction process.

The EMR is slow

Possible cause: Laboratory board load; busy hour; workstation.

Check: Whether only Laboratory is slow (known). Whether all screens are slow.

Do: Wait for the laboratory board. Try again. Tell Hospital Admin if the whole system is unusable.

Do not: Click submit repeatedly on payments or dispense.

Diagnoses unavailable / Coming soon on Nursing

Possible cause: Known limits of this release.

Do: Continue notes, vitals, and MAR. Ask the doctor for the diagnosis.

Do not: Invent a diagnosis or use Coming soon boxes as if they stored data.

When to contact Hospital Admin

- Nobody can sign in
- A staff account must be disabled
- A department is missing from someone who should have it
- The system is down
- Repeated errors after you confirmed identity and payment

What never to do

- Share accounts
- Change times
- Bypass payment
- Fabricate results
- Manipulate hidden records

System architecture

This document is for Hospital Admin, implementers, and technical support. Staff desk manuals avoid these names.

Release: `v1.0.0-uat`.

Related: Hospital deployment · Deployment secrets

Non-technical: how information moves

1. A staff member signs in to **Abikan** in the browser.
2. Abikan shows the **department** that person is allowed to use.
3. When they save clinical work (consultation, diagnosis, prescription, visit), it is stored in the **clinical chart**.
4. When they create a charge or collect payment, the **financial record** is stored in the billing system.
5. Laboratory **processing** (specimen, accession, result entry) happens in the laboratory system. Abikan only shows orders and released results.
6. Imaging **files**, if a scanner is connected, live in the imaging archive. Abikan does not show images in this release.
7. Ward notes, vitals, and MAR are recorded through Abikan and stored as clinical observations / administrations on the admission.

Abikan does **not** keep a second patient database. It does not invent a second invoice book.

Technical: request path

```
Staff browser
    → Bahmni proxy (ports 80 / 443)
        → Abikan UI (`/` and `/abikan/` from `abikan-shell/dist`)
        → Abikan API (`/openmrs/abikan-api` → abikan-api)
        → OpenMRS / Bahmni clinical apps
        → OpenELIS (`/openelis`)
        → other Bahmni paths (documents, IPD UI, and so on)

Abikan API
    → OpenMRS REST (and MySQL for admin analytics)
    → OpenELIS (server-side read of accession status only)
    → Odoo XML-RPC (charges, invoices, payments, stock)
```

System	Authoritative for
OpenMRS	Patient identity, visits, encounters, diagnoses, orders, observations, admissions, MAR administrations
OpenELIS	Specimen, accession, analysis, result validation
Odoo	Products, pricelists, sale orders, invoices, payments, pharmacy pickings / stock
dcm4chee / PACS	Imaging files when configured. **Not wired into Abikan screens in this release**
Abikan API + UI	Department workflow, RBAC for hospital jobs, department pricing writes, cashier, ward board helpers, schedule ensure, analytics, staff-admin audit
Bahmni proxy	Single staff entry, TLS, routing

Staff administration audit (who created or disabled accounts) is SQLite in Docker volume `abikan-admin-audit`.

Why there is no duplicate clinical database

Abikan reads and writes the existing engines. A nursing note is an observation on the admission encounter. A prescription is an OpenMRS drug order; the bill is an Odoo invoice. A laboratory result displayed in Abikan is the result already posted to OpenMRS after OpenELIS validation.

Observations and vitals

- **OPD vitals:** doctor saves with the consultation encounter on an open OPD visit.
- **Inpatient vitals:** nurse posts through Abikan; each save creates **new** OpenMRS observations on the admission encounter. Time and author are server-owned. No overwrite, no client timestamp.

MAR scheduling

Bahmni IPD does not always create slots when the doctor saves a drug order. On MAR load, Abikan asks the installed IPD schedule service to create schedules for eligible **inpatient** orders. PRN / as-needed orders are refused for the scheduled sheet. Administration uses the medication administration module; actual time is server clock.

Pharmacy

Atomfeed creates the commercial draft. Confirm / automated invoicing posts the invoice. Accountant payment is required. Pharmacist dispense validates the Odoo picking. Abikan does not write an OpenMRS “Dispensed” observation in this release.

Billing

See BILLING_AND_PAYMENTS.md. Catalogue `list_price` is copied onto the charge line at creation. Historical `account.move` lines are not rewritten.

Company currency may remain **USD** while departmental pricelists and new charges use **NGN**.

Docker services (summary)

Bahmni Standard compose plus Abikan overlay (`docker-compose.override.yml`). Hospital LAN hardening is `docker-compose.hospital.yml` (do not apply on the development Mac).

Hospital UI updates: `git checkout v1.0.0-uat` then `npm ci && npm run build`. Dist is generated, not tracked.

Never run `docker compose down -v` on a hospital host.